

Gaussian Process Emulators for Computer Experiments with Inequality Constraints

Hassan Maatouk¹ · Xavier Bay¹

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Abstract Physical phenomena are observed in many fields (science and engineering) and are often studied by time-consuming computer codes. These codes are analyzed with statistical models, often called emulators. In many situations, the physical system (computer model output) may be known to satisfy inequality constraints with respect to some or all input variables. The aim is to build a model capable of incorporating both data interpolation and inequality constraints into a Gaussian process emulator. By using a functional decomposition, a finite-dimensional approximation of Gaussian processes such that all conditional simulations satisfy the inequality constraints in the entire domain is proposed. To show the performance of the proposed model, some conditional simulations with inequality constraints such as boundedness, monotonicity or convexity conditions in one and two dimensions are given. A simulation study to investigate the efficiency of the method in terms of prediction is included.

Keywords Gaussian process emulator · Inequality constraints · Finite-dimensional approximation · Uncertainty quantification · Design and modeling of computer experiments

1 Introduction

In the engineering activity, runs of a computer code can be expensive and time-consuming. One solution is to use a statistical surrogate for conditioning computer

✉ Hassan Maatouk
hassan.maatouk@mines-stetienne.fr

Xavier Bay
bay@emse.fr

¹ École des Mines de Saint-Étienne, 158 Cours Fauriel, 42000 Saint-Étienne, France

model outputs at some input locations (design points). Gaussian process (GP) emulator is one of the most popular choices (Sacks et al. 1989). The reason comes from the property of the GP that uncertainty can be quantified. Furthermore, it has several nice properties. For example, the conditional GP at observation data (linear equality constraints) is still a GP (Cramer and Leadbetter 1967). Additionally, some inequality constraints (such as monotonicity and convexity) of output computer responses are related to partial derivatives. The partial derivatives of the GP are also Gaussian processes (GPs) (Cramer and Leadbetter 1967; Parzen 1962). Incorporating an infinite number of linear inequality constraints into a GP emulator, the problem becomes more difficult. The reason is that the resulting conditional process is not a GP in general.

In the literature of interpolation with inequality constraints, two types of methods are found. The first one is deterministic and based on splines, which has the advantage that inequality constraints are satisfied in the entire domain (Fritsch and Carlson 1980; Micchelli and Utreras 1988; Villalobos and Wahba 1987; Wolberg and Alf 2002; Wright and Wegman 1980). The second one is based on the simulation of the conditional GP by using the subdivision of the input set (Abrahamsen and Benth 2001; Da Veiga and Marrel 2012; Golchi et al. 2015; Riihimäki and Vehtari 2010; Xiaoqing 2012). In that case, the inequality constraints are satisfied in a finite number of input locations. However, uncertainty can be quantified. In this framework, constrained kriging has been studied in the domain of geostatistics (Freulon and de Fouquet 1993; Kleijnen and Beers 2013). In previous work, some methodologies have been based on the knowledge of the derivatives of the GP at some input locations (Golchi et al. 2015; Riihimäki and Vehtari 2010; Xiaoqing 2012). For monotonicity constraints with noisy data, a Bayesian approach was developed in Riihimäki and Vehtari (2010). In Golchi et al. (2015), the problem is to build a GP emulator by using the prior monotonicity information of the computer model response with respect to some inputs. Their idea is based on an approach similar to Riihimäki and Vehtari (2010) placing the derivatives information at specified input locations, by forcing the derivative process to be positive at these points. In such methodology, monotonicity constraints are not guaranteed in the entire domain. Recently, a methodology based on a discrete location approximation for incorporating inequality constraints into a GP emulator was developed in Da Veiga and Marrel (2012). Again, the inequality constraints are not guaranteed in the entire domain.

Villalobos and Wahba (1987) used splines to estimate an interpolation smooth function satisfying a finite number of linear inequality constraints. In terms of estimation of monotone smoothing functions, using B-splines was firstly introduced by Ramsay (1988). The idea is based on the integration of B-splines defined on a properly set of knots with positive coefficients to ensure monotonicity constraints. A similar approach is applied to econometrics in Dole (1999). Xuming and Peide (1996) take the same approach and suggest the calculation of the coefficients by solving a finite linear minimization problem. A comparison to monotone kernel regression and an application to decreasing constraints are included.

The aim of this paper is to build a GP emulator incorporating the advantage of splines approach in order to ensure that inequality constraints are satisfied in the entire domain. A finite-dimensional approximation of Gaussian processes that converges uniformly pathwise is proposed. It is constructed by a linear decomposition of deterministic basis

functions with Gaussian random coefficients, where these coefficients are not independent. The basis functions can be chosen such that inequality constraints of the GP are equivalent to constraints on the coefficients. Therefore, the inequality constraints are reduced to a finite number of constraints. Furthermore, any posterior sample of coefficients leads to an interpolating function satisfying the inequality constraints in the entire domain. Finally, the problem is reduced to simulate a Gaussian vector (random coefficients) restricted to convex sets which is a well-known problem with existing algorithms (Botts 2013; Chopin 2011; Emery et al. 2014; Freulon and de Fouquet 1993; Geweke 1991; Maatouk and Bay 2016; Philippe and Robert 2003; Robert 1995).

The article is structured as follows: In Sect. 2, Gaussian process modeling for computer experiments and the choice of covariance functions are briefly recalled. In Sect. 3, a finite-dimensional approximation of GPs capable of interpolating computer model outputs and incorporating inequality constraints in the entire domain is proposed, and its properties are investigated. In Sect. 4, the performance of the proposed model in terms of prediction is investigated. In Sect. 5, some simulated examples of the conditional GP with inequality constraints (such as boundedness, monotonicity or convexity conditions) in one and two dimensions are shown. Moreover, two cases of truncated simulations are studied followed by some concluding remarks and future work at the end of the paper.

2 Gaussian Process Emulators for Computer Experiments

The model $y = f(x)$ is considered, where the simulator response y is assumed to be a deterministic real-valued function of the d -dimensional variable $x = (x_1, \dots, x_d) \in \mathbb{R}^d$. The real function is supposed to be continuous and evaluated at n design points given by the rows of the $n \times d$ matrix $X = (x^{(1)}, \dots, x^{(n)})^\top$, where $x^{(i)} \in \mathbb{R}^d$, $1 \leq i \leq n$. In practice, the evaluation of the function is expensive and must be considered highly time-consuming. The solution is to estimate the unknown function f by using a GP emulator also known as ‘kriging.’ In this framework, y is viewed as a realization of a continuous GP

$$Y(x) = \eta(x) + Z(x),$$

where the deterministic continuous function $\eta : x \in \mathbb{R}^d \rightarrow \eta(x) \in R$ is the mean and Z is a zero-mean GP with continuous covariance function

$$K : (u, v) \in \mathbb{R}^d \times \mathbb{R}^d \rightarrow K(u, v) \in \mathbb{R}.$$

Conditionally to the observation $y = (y(x^{(1)}), \dots, y(x^{(n)}))^\top$, the process is still a GP

$$Y(x) \mid Y(X) = y \sim \mathcal{N}(\zeta(x), \tau^2(x)), \quad (1)$$

where

$$\begin{aligned} \zeta(x) &= \eta(x) + k(x)^\top \mathbb{K}^{-1} (y - \mu) \\ \tau^2(x) &= K(x, x) - k(x)^\top \mathbb{K}^{-1} k(x), \end{aligned}$$

Table 1 Some popular covariance functions used in Kriging methods

Name	Expression	Class
Gaussian	$\sigma^2 \exp\left(-\frac{(x-x')^2}{2\theta^2}\right)$	$\mathcal{C}^\infty$
Matérn 5/2	$\sigma^2 \left(1 + \frac{\sqrt{5} x-x' }{\theta} + \frac{5(x-x')^2}{3\theta^2}\right) \exp\left(-\frac{\sqrt{5} x-x' }{\theta}\right)$	$\mathcal{C}^2$
Matérn 3/2	$\sigma^2 \left(1 + \frac{\sqrt{3} x-x' }{\theta}\right) \exp\left(-\frac{\sqrt{3} x-x' }{\theta}\right)$	$\mathcal{C}^1$
Exponential	$\sigma^2 \exp\left(-\frac{ x-x' }{\theta}\right)$	$\mathcal{C}^0$

and $\mu = \eta(X)$ is the vector of trend values at the experimental design points, $\mathbb{K}_{i,j} = K(x^{(i)}, x^{(j)})$, $i, j = 1, \dots, n$ is the covariance matrix of $Y(X)$ and $k(x) = K(x, x^{(i)})$ is the vector of covariance between $Y(x)$ and $Y(X)$. Additionally, the covariance function between any two inputs is that

$$C(x, x') = \text{Cov}(Y(x), Y(x') \mid Y(X) = y) = K(x, x') - k(x)^\top \mathbb{K}^{-1} k(x'),$$

where C is the covariance function of the conditional GP. The mean $\zeta(x)$ is called kriging mean prediction of $Y(x)$ based on the computer model outputs $Y(X) = y$ (Jones et al. 1998).

2.1 The Choice of Covariance Function

The choice of K has crucial consequences specially in controlling the smoothness of the kriging metamodel. It must be chosen in the set of definite and positive kernels. Some popular covariance functions are the Gaussian covariance function, Matérn covariance function (with parameter $\lambda = 3/2, 5/2, \dots$) and exponential covariance function (Matérn covariance function with parameter $\lambda = 1/2$). Notice that these covariance function are placed in decreasing order of smoothness, the Gaussian covariance function corresponding to $\mathcal{C}^\infty$ function (i.e., the space of functions that admit derivatives of all orders) and the exponential covariance function to continuous one (Rasmussen and Williams 2005 and Table 1). In the running examples of this paper, the Gaussian covariance function defined by

$$K(x, x') = \sigma^2 \prod_{k=1}^d \exp\left(-\frac{(x_k - x'_k)^2}{2\theta_k^2}\right),$$

for all $x, x' \in \mathbb{R}^d$ will be considered, where σ^2 and $\theta = (\theta_1, \dots, \theta_d)$ are parameters.

2.2 Derivatives of Gaussian Processes

In this paragraph, the paths of $Y(x)$ are assumed to be of class $\mathcal{C}^p$ (i.e., the space of functions that admit derivatives up to order p). This can be guaranteed if K is smooth enough, and in particular if K is of class $\mathcal{C}^\infty$ (Cramer and Leadbetter 1967). The linearity of the differentiation operation ensures that the order partial derivatives of a GP are also GPs (Cramer and Leadbetter 1967), with (Parzen 1962)

$$\begin{aligned}\mathbb{E} \left(\partial_{x_k}^p Y(x) \right) &= \frac{\partial^p}{\partial x_k^p} \eta(x), \\ \text{Cov} \left(\partial_{x_k}^p Y \left(x^{(i)} \right), \partial_{x_\ell}^q Y \left(x^{(j)} \right) \right) &= \frac{\partial^{p+q}}{\partial x_k^p \partial (x'_\ell)^q} K \left(x^{(i)}, x^{(j)} \right).\end{aligned}$$

3 Gaussian Process Emulators with Inequality Constraints

In this section, the real function (physical system) is assumed to satisfy inequality constraints (such as boundedness, monotonicity or convexity conditions) in the entire domain. The aim is to incorporate both interpolation conditions and inequality constraints into a Gaussian process emulator.

3.1 Formulation of the Problem

Without loss of generality, the input x is in $[0, 1]^d \subset \mathbb{R}^d$. The real function f is evaluated at n distinct locations in the input set

$$f \left(x^{(i)} \right) = y_i, \quad i = 1, \dots, n.$$

Let $(Y(x))_{x \in [0, 1]^d}$ be a zero-mean GP with covariance function K and $\mathcal{C}^0([0, 1]^d)$ be the space of continuous function on $[0, 1]^d$. Let C be the subset of $\mathcal{C}^0([0, 1]^d)$ that corresponding to a given set of linear inequality constraints. The aim is to get the conditional distribution of Y given interpolation conditions and inequality constraints, respectively, as

$$\begin{aligned}Y \left(x^{(i)} \right) &= y_i, \quad i = 1, \dots, n, \\ Y &\in C.\end{aligned}$$

3.2 Gaussian Process Approximation

To handle the conditional distribution incorporating both interpolation conditions and inequality constraints, a finite-dimensional approximation Y^N of Gaussian processes Y of the form

$$Y^N(x) = \sum_{j=0}^N \xi_j \phi_j(x), \quad x \in \mathbb{R}^d, \quad (2)$$

is proposed, where $\xi = (\xi_0, \dots, \xi_N)^\top$ is a zero-mean Gaussian vector with covariance matrix Γ^N and $\phi = (\phi_0, \dots, \phi_N)^\top$ is a vector of basis functions. The choice of these basis functions and Γ^N depends on the type of inequality constraints. Notice that Y^N is a zero-mean GP with covariance function

$$K_N(x, x') = \phi(x)^\top \Gamma^N \phi(x').$$

This type of covariance functions has been used in [Cressie and Johannesson \(2008, Eq. 2.12\)](#), where the basis functions are not necessary orthogonal and Γ^N is a square positive definite matrix estimated from the data. This is not the case in the present paper. In the following sections, some examples of the choice of the basis functions are given and the covariance matrix Γ^N of the Gaussian vector ξ is computed such that the finite-dimensional approximation Y^N converges to the original GP Y . The advantage of the proposed Model (2) is that the simulation of the conditional GP is reduced to the simulation of the Gaussian vector ξ given that

$$\sum_{j=0}^N \xi_j \phi_j(x^{(i)}) = y_i, \quad i = 1, \dots, n, \quad (3)$$

$$\xi \in C_\xi, \quad (4)$$

where $C_\xi = \{c \in \mathbb{R}^{N+1} : \sum_{j=0}^N c_j \phi_j \in C, c = (c_0, \dots, c_N)^\top\}$. Hence, the problem is equivalent to simulate a Gaussian vector restricted to (3) and (4).

3.3 One-Dimensional Cases

3.3.1 Boundedness Constraints

The real function defined in the unit interval is supposed to be continuous ($a \leq f(x) \leq b$, $x \in [0, 1]$, where $-\infty \leq a < b \leq +\infty$). In that case, the convex set C is the space of bounded functions and is defined as

$$C = \left\{ f \in \mathcal{C}^0([0, 1]) : a \leq f(x) \leq b, x \in [0, 1] \right\}.$$

Let us begin by constructing the functions h_j , $j = 0, \dots, N$ that will be used in the proposed model. By discretizing the input set as $0 = u_0 < u_1 < \dots < u_N = 1$, one function is build on each knot. For the sake of simplicity, a uniform subdivision of the input set is used, but the methodology can be adapted for any subdivision. For example, at the j th knot $u_j = j \Delta_N = j/N$, the associated function is

$$h_j(x) = h\left(\frac{x - u_j}{\Delta_N}\right), \quad j = 0, \dots, N, \quad (5)$$

where $\Delta_N = 1/N$ and $h(x) = (1 - |x|) \mathbb{1}_{(|x| \leq 1)}$, $x \in \mathbb{R}$. Notice that the h_j 's are bounded between zero and one and $\sum_{j=0}^N h_j(x) = 1$ for all x in $[0, 1]$. Additionally,

the value of these functions at any knot u_i , $i = 0, \dots, N$ is equal to the Kronecker's delta ($h_j(u_i) = \delta_{ij}$), where δ_{ij} is equal to one if $i = j$ and zero otherwise.

The philosophy of the proposed method is presented in the following proposition:

Proposition 1 *With the notations introduced before, the finite-dimensional approximation of GPs $(Y^N(x))_{x \in [0,1]}$ is defined as*

$$Y^N(x) = \sum_{j=0}^N Y(u_j) h_j(x) = \sum_{j=0}^N \xi_j h_j(x), \quad (6)$$

where $\xi_j = Y(u_j)$, $j = 0, \dots, N$. If the realizations of the original GP Y are continuous, then the following properties hold:

- Y^N is a finite-dimensional GP (i.e., a GP with paths lying in a finite-dimensional space is called a finite-dimensional GP) with covariance function $K_N(x, x') = h(x)^\top \Gamma^N h(x')$, where $h(x) = (h_0(x), \dots, h_N(x))^\top$, $\Gamma_{i,j}^N = K(u_i, u_j)$, $i, j = 0, \dots, N$ and K the covariance function of the original GP Y .
- Y^N converges uniformly pathwise to Y when N tends to infinity. This means that $Y^N(x; \omega)$ is uniformly convergent as a function of x to $Y(x; \omega)$ for each realization ω .
- Y^N is in C if and only if the $(N + 1)$ coefficients $Y(u_j)$ are contained in $[a, b]$.

The advantage of this model is that the infinite number of inequality constraints of Y^N are equivalent to a finite number of constraints on the coefficients $(Y(u_j))_{0 \leq j \leq N}$. Therefore, the problem is reduced to simulate the Gaussian vector $\xi = (Y(u_0), \dots, Y(u_N))^\top$ restricted to the convex subset formed by the two constraints (3) and (4), where $C_\xi = \{\xi \in \mathbb{R}^{N+1} : a \leq \xi_j \leq b, j = 0, \dots, N\}$. The proof of this proposition is available in “Appendix.”

Simulated Paths As shown in Proposition 1, the simulation of the finite-dimensional approximation of Gaussian processes Y^N conditionally to given data and boundedness constraints ($Y^N \in I \cap C$) is reduced to simulate the Gaussian vector ξ restricted to $I_\xi \cap C_\xi$

$$I_\xi = \left\{ \xi \in \mathbb{R}^{N+1} : A\xi = y \right\},$$

$$C_\xi = \left\{ \xi \in \mathbb{R}^{N+1} : a \leq \xi_j \leq b, j = 0, \dots, N \right\},$$

where the $n \times (N + 1)$ matrix A is defined as $A_{i,j} = h_j(x^{(i)})$. The interpolation system $A\xi = y$ admits solutions only if $N + 1 - n \geq 1$ (number of degrees of freedom).

The sampling scheme can be summarized in two steps: First, the conditional distribution of the Gaussian vector ξ with respect to data interpolation is computed

$$\xi \mid A\xi = y \sim \mathcal{N} \left(\left((A\Gamma^N)^\top (A\Gamma^N A^\top)^{-1} y, \Gamma^N - (A\Gamma^N)^\top (A\Gamma^N A^\top)^{-1} A\Gamma^N \right). \right. \quad (7)$$

Then, the Gaussian vector ξ with the above distribution (7) is simulated using an improved rejection sampling (Maatouk and Bay 2016) in which only the random coefficients in the convex set $[a, b]$ are selected. The sample paths of the conditional Gaussian process are generated by Eq. (6) and hence satisfy both interpolation conditions and boundedness constraints in the entire domain (R package developed in Maatouk and Richet 2015).

3.3.2 Monotonicity Constraints

In this section, the real function f is assumed to be of class $\mathcal{C}^1$. The convex set C is the space of non-decreasing functions and is defined as

$$C = \left\{ f \in \mathcal{C}^1([0, 1]) : f'(x) \geq 0, x \in [0, 1] \right\}.$$

Since the monotonicity is related to the sign of the derivative, then the proposed model is adapted from Model (6). The basis functions are defined as the primitive functions of h_j

$$\phi_j(x) = \int_0^x h_j(t) dt, \quad x \in [0, 1].$$

Remark that the derivative of the basis functions ϕ_j at any knot u_i , $i = 0, \dots, N$ is equal to the Kronecker's delta ($\phi_j'(u_i) = \delta_{ij}$).

Similarly to Proposition 1, the following results hold:

Proposition 2 *Suppose that the realizations of the original GP Y are almost surely continuously differentiable. Using the notations introduced before, the finite-dimensional approximation of Gaussian processes $(Y^N(x))_{x \in [0, 1]}$ is defined as*

$$Y^N(x) = Y(0) + \sum_{j=0}^N Y'(u_j) \phi_j(x) = \zeta + \sum_{j=0}^N \xi_j \phi_j(x), \quad (8)$$

where $\zeta = Y(0)$ and $\xi_j = Y'(u_j)$, $j = 0, \dots, N$. Then, the following properties hold:

- Y^N is a finite-dimensional GP with covariance function

$$K_N(x, x') = \left(1, \phi(x)^\top \right) \Gamma_{\text{new}}^N \left(1, \phi(x')^\top \right)^\top,$$

where $\phi(x) = (\phi_0(x), \dots, \phi_N(x))^\top$ and Γ_{new}^N is the covariance matrix of the Gaussian vector $(\zeta, \xi) = (Y(0), Y'(u_0), \dots, Y'(u_N))^\top$ which is equal to

$$\Gamma_{\text{new}}^N = \begin{bmatrix} K(0, 0) & \frac{\partial K}{\partial x'}(0, u_j) \\ \frac{\partial K}{\partial x}(u_i, 0) & \Gamma_{i,j}^N \end{bmatrix}_{0 \leq i, j \leq N},$$

with $\Gamma_{i,j}^N = \frac{\partial^2 K}{\partial x \partial x'}(u_i, u_j)$, $i, j = 0, \dots, N$ and K the covariance function of the original GP Y .

- Y^N converges uniformly to Y when N tends to infinity (with probability one).
- Y^N is non-decreasing if and only if the coefficients $(Y'(u_j))_{0 \leq j \leq N}$ are all non-negative.

The proof of this proposition is available in “Appendix.” From the last property, the problem is reduced to simulate the Gaussian vector (ζ, ξ) restricted to the convex set formed by the interpolation conditions and the inequality constraints, respectively,

$$Y^N(x^{(i)}) = \zeta + \sum_{j=0}^N \xi_j \phi_j(x^{(i)}) = y_i, \quad i = 1, \dots, n,$$

$$(\zeta, \xi) \in C_\xi = \left\{ (\zeta, \xi) \in \mathbb{R}^{N+2} : \xi_j \geq 0, j = 0, \dots, N \right\}.$$

Simulated Paths As shown in Proposition 2, the simulation of the finite-dimensional approximation of Gaussian processes Y^N conditionally to given data and monotonicity constraints ($Y^N \in I \cap C$) is reduced to simulate the Gaussian vector (ζ, ξ) restricted to $I_\xi \cap C_\xi$

$$I_\xi = \left\{ (\zeta, \xi) \in \mathbb{R}^{N+2} : A(\zeta, \xi) = y \right\},$$

$$C_\xi = \left\{ (\zeta, \xi) \in \mathbb{R}^{N+2} : \xi_j \geq 0, j = 0, \dots, N \right\},$$

where the $n \times (N+2)$ matrix A is defined as

$$A_{i,j} = \begin{cases} 1 & \text{for } i = 1, \dots, n \text{ and } j = 1, \\ \phi_{j-2}(x^{(i)}) & \text{for } i = 1, \dots, n \text{ and } j = 2, \dots, N+2. \end{cases}$$

The Gaussian vector (ζ, ξ) with the conditional distribution defined in (7) is simulated, where Γ^N is replaced by Γ_{new}^N . Then, using an improved rejection sampling (Maa-touk and Bay 2016), the nonnegative coefficients ξ_j are selected. Finally, the sample paths of the conditional Gaussian process are generated by Eq. (8) which satisfy both interpolation conditions and monotonicity constraints in the entire domain.

Remark 1 (Monotonicity of continuous but non-derivable functions) If the real function is of class $\mathcal{C}^0$ only (but possibly not derivable) and non-decreasing in the entire domain, then the proposed model defined in (6) is non-decreasing if and only if the sequence of coefficients $(Y(u_j))_j$, $j = 0, \dots, N$ is non-decreasing (i.e., $Y(u_{j-1}) \leq Y(u_j)$, $j = 1, \dots, N$). The simulated paths are generated using the same strategy in Sect. 3.3.1, where $C_\xi = \{\xi \in \mathbb{R}^{N+1} : \xi_{j-1} \leq \xi_j, j = 1, \dots, N\}$.

3.3.3 Convexity Constraints

In this section, the real function is supposed to be twice differentiable. Since the functions h_j , $j = 0, \dots, N$ defined in (5) are all nonnegative, then the basis functions

φ_j are taken as the two times primitive functions of h_j

$$\varphi_j(x) = \int_0^x \left(\int_0^t h_j(u) du \right) dt.$$

Similarly to the monotonicity case, the second derivative of the basis functions φ_j'' at any knot u_i , ($0 \leq i \leq N$) is equal to Kronecker's delta ($\varphi_j''(u_i) = \delta_{ij}$). The realizations of the original GP Y are assumed to be at least twice differentiable. The finite-dimensional approximation defined as

$$Y^N(x) = Y(0) + Y'(0)x + \sum_{j=0}^N Y''(u_j)\varphi_j(x) = \zeta + \kappa x + \sum_{j=0}^N \xi_j \varphi_j(x), \quad (9)$$

is convex if and only if the $(N+1)$ random coefficients $\xi_j = Y''(u_j)$ are all non-negative, where $\zeta = Y(0)$ and $\kappa = Y'(0)$. Thus, the problem is reduced to generate the Gaussian vector $(\zeta, \kappa, \xi) = (Y(0), Y'(0), Y''(u_0), \dots, Y''(u_N))^\top$ restricted to the convex set $I_\xi \cap C_\xi$, where

$$I_\xi = \left\{ (\zeta, \kappa, \xi) \in \mathbb{R}^{N+3} : \zeta + \kappa x^{(i)} + \sum_{j=0}^N \xi_j \varphi_j(x^{(i)}) = y_i, i = 1, \dots, n \right\},$$

$$C_\xi = \left\{ (\zeta, \kappa, \xi) \in \mathbb{R}^{N+3} : \xi_j \geq 0, j = 0, \dots, N \right\}.$$

Its $(N+3) \times (N+3)$ covariance matrix Γ_{new}^N is given by

$$\Gamma_{\text{new}}^N = \begin{bmatrix} \text{Var}(\zeta) & \text{Cov}(\zeta, \kappa) & \text{Cov}(\zeta, \xi) \\ \text{Cov}(\kappa, \zeta) & \text{Var}(\kappa) & \text{Cov}(\kappa, \xi) \\ \text{Cov}(\xi, \zeta) & \text{Cov}(\xi, \kappa) & \text{Cov}(\xi, \xi) \end{bmatrix} = \begin{bmatrix} K(0, 0) & \frac{\partial K}{\partial x'}(0, 0) & \frac{\partial^2 K}{\partial (x')^2}(0, u_j) \\ \frac{\partial K}{\partial x}(0, 0) & \frac{\partial^2 K}{\partial x \partial x'}(0, 0) & \frac{\partial^3 K}{\partial x \partial (x')^2}(0, u_j) \\ \frac{\partial^2 K}{\partial x^2}(u_i, 0) & \frac{\partial^3 K}{\partial x^2 \partial x'}(u_i, 0) & \Gamma_{i,j}^N \end{bmatrix}_{0 \leq i, j \leq N},$$

where

$$\Gamma_{i,j}^N = \text{Cov}(\xi_i, \xi_j) = \text{Cov}(Y''(u_i), Y''(u_j)) = \frac{\partial^4 K}{\partial x^2 \partial (x')^2}(u_i, u_j), \quad i, j = 0, \dots, N.$$

Finally, the covariance function of the finite-dimensional approximation of GPs is equal to

$$K_N(x, x') = \left(1, x, \varphi(x)^\top\right) \Gamma_{\text{new}}^N \left(1, x', \varphi(x')^\top\right)^\top,$$

where $\varphi(x) = (\varphi_0(x), \dots, \varphi_N(x))^\top$.

Simulated Paths As shown in this section, the simulation of the finite-dimensional approximation of Gaussian processes Y^N conditionally to given data and convexity constraints ($Y^N \in I \cap C$) is reduced to simulate the Gaussian vector (ζ, κ, ξ) restricted to $I_\xi \cap C_\xi$

$$I_\xi = \left\{ (\zeta, \kappa, \xi) \in \mathbb{R}^{N+3} : A(\zeta, \kappa, \xi) = y \right\},$$

$$C_\xi = \left\{ (\zeta, \kappa, \xi) \in \mathbb{R}^{N+3} : \xi_j \geq 0, \quad j = 0, \dots, N \right\},$$

where the $n \times (N + 3)$ matrix A is

$$A_{i,j} = \begin{cases} 1 & \text{for } i = 1, \dots, n \text{ and } j = 1, \\ x^{(i)} & \text{for } i = 1, \dots, n \text{ and } j = 2, \\ \varphi_{j-3}(x^{(i)}) & \text{for } i = 1, \dots, n \text{ and } j = 3, \dots, N + 3. \end{cases}$$

The Gaussian vector (ζ, κ, ξ) is simulated with the conditional distribution defined in (7), where Γ^N is replaced by Γ_{new}^N . Then, using an improved rejection sampling (Maatouk and Bay 2016), the nonnegative coefficients ξ_j are selected. Finally, the sample paths of the conditional Gaussian process are generated by Eq. (9) which satisfy both interpolation conditions and convexity constraints in the entire domain.

The problem dimension $d \geq 2$ is considered. For boundedness constraints, the proposed model can be easily extended to multidimensional cases. In the following, isotonicity constraints are studied.

3.4 Isotonicity in Two Dimensions

The input is $x = (x_1, x_2) \in \mathbb{R}^2$ and without loss of generality is assumed in the unit square. The real function f is supposed to be monotone (non-decreasing for example) with respect to the two input variables

$$x_1 \leq x'_1 \quad \text{and} \quad x_2 \leq x'_2 \quad \Rightarrow \quad f(x_1, x_2) \leq f(x'_1, x'_2).$$

The idea is the same as in the one-dimensional case. The basis functions are constructed such that monotonicity constraints are equivalent to constraints on the coefficients. Firstly, the unit square is discretized (e.g., uniformly to $(N + 1)^2$ knots; see Fig. 6 for $N = 7$). Secondly, on each knot one basis function is built. For instance, the basis function at the knot (u_i, u_j) is

$$\Phi_{i,j}(x) = h_i(x_1)h_j(x_2),$$

where h_j , $j = 0, \dots, N$ are defined in (5). Then

$$\Phi_{i,j}(u_k, u_\ell) = \delta_{i,k}\delta_{j,\ell}, \quad k, \ell = 0, \dots, N.$$

Proposition 3 Using the notations introduced before, the finite-dimensional approximation of Gaussian processes $(Y^N(x))_{x \in [0,1]^2}$ is defined as

$$Y^N(x_1, x_2) = \sum_{i,j=0}^N Y(u_i, u_j) h_i(x_1) h_j(x_2) = \sum_{i,j=0}^N \xi_{i,j} h_i(x_1) h_j(x_2), \quad (10)$$

where $\xi_{i,j} = Y(u_i, u_j)$ and the functions h_j , $j = 0, \dots, N$ are defined in (5). Then, the following properties hold:

- Y^N is a finite-dimensional GP with covariance function $K_N(x, x') = \Phi(x)^\top \Gamma^N \Phi(x')$, where $\Phi(x)^\top = (h_i(x_1) h_j(x_2))_{i,j}$, $\Gamma_{(i,j),(i',j')}^N = K((u_i, u_j), (u_{i'}, u_{j'}))$ and K is the covariance function of the original GP Y .
- Y^N converges uniformly to Y when N tends to infinity (with probability one).
- Y^N is non-decreasing with respect to the two input variables if and only if the $(N+1)^2$ random coefficients $\xi_{i,j}$, $i, j = 0, \dots, N$ verify the following linear constraints:
 1. $\xi_{i-1,j} \leq \xi_{i,j}$ and $\xi_{i,j-1} \leq \xi_{i,j}$, $i, j = 1, \dots, N$.
 2. $\xi_{i-1,0} \leq \xi_{i,0}$, $i = 1, \dots, N$.
 3. $\xi_{0,j-1} \leq \xi_{0,j}$, $j = 1, \dots, N$.

The proof of this proposition is similar to that in the one-dimensional case. From the last property, the problem is reduced to simulate the Gaussian vector $\xi = (\xi_{i,j})_{i,j}$ restricted to the convex set $I_\xi \cap C_\xi$, where

$$I_\xi = \left\{ \xi \in \mathbb{R}^{(N+1)^2} : Y^N(x_1^{(i)}, x_2^{(i)}) = \sum_{i,j=0}^N \xi_{i,j} h_i(x_1^{(i)}) h_j(x_2^{(i)}) = y_i \right\},$$

$$C_\xi = \left\{ \xi \in \mathbb{R}^{(N+1)^2} \text{ such that } \xi_{i,j} \text{ verify the constraints 1. 2. and 3.} \right\}.$$

Remark 2 (Isotonicity in two dimensions with respect to one variable) If the function is non-decreasing with respect to the first variable only, then the proposed GP defined as

$$Y^N(x_1, x_2) = \sum_{i,j=0}^N Y(u_i, u_j) h_i(x_1) h_j(x_2) = \sum_{i,j=0}^N \xi_{i,j} h_i(x_1) h_j(x_2), \quad (11)$$

is non-decreasing with respect to x_1 if and only if the random coefficients $\xi_{i-1,j} \leq \xi_{i,j}$, $i = 1, \dots, N$ and $j = 0, \dots, N$.

3.5 Isotonicity in Multidimensional Cases

The d -dimensional case is a simple extension of the two-dimensional case. The finite-dimensional approximation of Gaussian processes Y^N is that

$$\begin{aligned}
 Y^N(x) &= \sum_{i_1, \dots, i_d=0}^N Y(u_{i_1}, \dots, u_{i_d}) \prod_{\sigma \in \{1, \dots, d\}} h_{i_\sigma}(x_\sigma) \\
 &= \sum_{i_1, \dots, i_d=0}^N \xi_{i_1, \dots, i_d} \prod_{\sigma \in \{1, \dots, d\}} h_{i_\sigma}(x_\sigma),
 \end{aligned}$$

where $\xi_{i_1, \dots, i_d} = Y(u_{i_1}, \dots, u_{i_d})$. Remark 2 can be extended as well, for the case of a monotonicity with respect to a subset of variables. For instance, the monotonicity of Y^N with respect to the ℓ -th-dimensional input x_ℓ is equivalent to the fact that $\xi_{i_1, \dots, i_{\ell-1}, \dots, i_d} \leq \xi_{i_1, \dots, i_\ell, \dots, i_d}$, $i_\ell = 1, \dots, N$ and $i_1, \dots, i_{\ell-1}, i_{\ell+1}, \dots, i_d = 0, \dots, N$.

3.6 Simulation of GPs Conditionally to Equality and Inequality Constraints

For the sake of simplicity and without loss of generality, the proposed finite-dimensional approximation of GPs is supposed of the form

$$Y^N(x) = \sum_{j=0}^N \xi_j \phi_j(x), \quad x \in \mathbb{R}^d,$$

where $\xi = (\xi_0, \dots, \xi_N)^\top$ is a zero-mean Gaussian vector with covariance matrix Γ^N and $\phi = (\phi_0, \dots, \phi_N)^\top$ are deterministic basis functions. For instance, the constant term $Y(0)$ in Model (8) can be written as $\xi_0 \phi_0(x)$, where $\phi_0(x) = 1$. The space of interpolation conditions is $I_\xi = \{\xi \in \mathbb{R}^{N+1} : \sum_{j=0}^N \xi_j \phi_j(x^{(i)}) = y_i, i = 1, \dots, n\}$ and the set of inequality constraints C_ξ is a convex set (for instance, the nonnegative quadrant $\xi_j \geq 0, j = 0, \dots, N$ for non-decreasing constraints in one dimension). The main interest is to calculate the mean, mode (maximum a posteriori) of Y^N conditionally to $\xi \in I_\xi \cap C_\xi$ and the quantification of prediction intervals. Note that their analytical forms except for the mode are not easy to find, hence the need of simulation. As explained in Sect. 3.3.1, the problem is reduced to simulate the Gaussian vector $\xi = (\xi_0, \dots, \xi_N)^\top$ restricted to convex sets. In that case, several algorithms can be used (Botts 2013; Chopin 2011; Geweke 1991; Kotecha and Djuric 1999; Maatouk and Bay 2016; Philippe and Robert 2003; Robert 1995).

In this section, some notations that will be used in Sect. 5 are introduced, and the two cases of truncated simulations are emphasized. Let us define ξ_1 as the mean of ξ conditionally to $\xi \in I_\xi$ without inequality constraints (Eq. 7). Then by linearity of the conditional expectation, the so-called usual (unconstrained) kriging mean is equal to

$$m_K^N(x) = \mathbb{E}\left(Y^N(x) \mid Y^N(x^{(i)}) = y_i, i = 1, \dots, n\right) = \sum_{j=0}^N (\xi_1)_j \phi_j(x),$$

where $\xi_1 = \mathbb{E}(\xi \mid \xi \in I_\xi) = \Gamma^N A^\top (A \Gamma^N A^\top)^{-1} y \in \mathbb{R}^{N+1}$ and the matrix A is formed by the values of the basis functions at the observations (i.e., $A_{i,j} = \phi_j(x^{(i)})$).

Similarly to the kriging mean of the original GP (Eq. (1)), the kriging mean m_K^N of the finite-dimensional approximation of GPs Y^N is

$$m_K^N(x) = k_N(x)^\top \mathbb{K}_N^{-1} y,$$

where $k_N(x) = (K_N(x, x^{(i)}))_i = (A\Gamma^N\phi(x))$ is the vector of covariance between $Y^N(x)$ and $Y^N(X)$ and $(\mathbb{K}_N)_{i,j} = K_N(x^{(i)}, x^{(j)})_{i,j} = (A\Gamma^N A^\top)_{i,j}$, $i, j = 1, \dots, n$ is the covariance matrix of $Y^N(X) = y$.

Definition 1 Denote ξ_C as the mean of the Gaussian vector ξ restricted to $I_\xi \cap C_\xi$ (i.e., the posterior mean). Then, the inequality kriging mean (mean a posteriori) is defined

$$m_{IK}^N(x) = \mathbb{E} \left(Y^N(x) \mid Y^N \left(x^{(i)} \right) = y_i, \xi \in C_\xi \right) = \sum_{j=0}^N (\xi_C)_j \phi_j(x),$$

where $\xi_C = \mathbb{E}(\xi \mid \xi \in I_\xi \cap C_\xi)$.

Finally, let μ be the maximum of the probability density function (pdf) of ξ restricted to $I_\xi \cap C_\xi$. It is the solution of the following convex optimization problem

$$\mu = \arg \min_{c \in I_\xi \cap C_\xi} \left(\frac{1}{2} c^\top \left(\Gamma^N \right)^{-1} c \right), \quad (12)$$

where Γ^N is the covariance matrix of the Gaussian vector ξ . In fact, μ corresponds to the mode (i.e., the maximum of the probability density function) of the Gaussian vector ξ restricted to $I_\xi \cap C_\xi$ and its numerical calculation is a standard problem in the minimization of positive quadratic forms subject to convex constraints (Boyd and Vandenberghe 2004; Goldfarb and Idnani 1983). Let us mention that in all simulation examples illustrated in this paper, the R Package ‘solve.QP’ described in Goldfarb and Idnani (1982, 1983) is used to compute the mode of the truncated Gaussian vector (i.e., to solve the quadratic convex optimization problem (12)).

Definition 2 The so-called inequality mode or maximum a posteriori (MAP) of the finite-dimensional approximation of GPs Y^N conditionally to given data and inequality constraints is equal to

$$M_{IK}^N(x) = \sum_{j=0}^N \mu_j \phi_j(x), \quad x \in \mathbb{R}^d,$$

where $\mu = (\mu_0, \dots, \mu_N)^\top$ is defined in (12).

Remark 3 The inequality mode M_{IK}^N defined in Definition 2 does not depend on the variance parameter σ of the covariance function K since the vector μ and the basis functions ϕ_j do not depend on it as well. Also, it does not depend on the simulation but on the length hyper-parameters of the covariance function $\theta = (\theta_1, \dots, \theta_d)$.

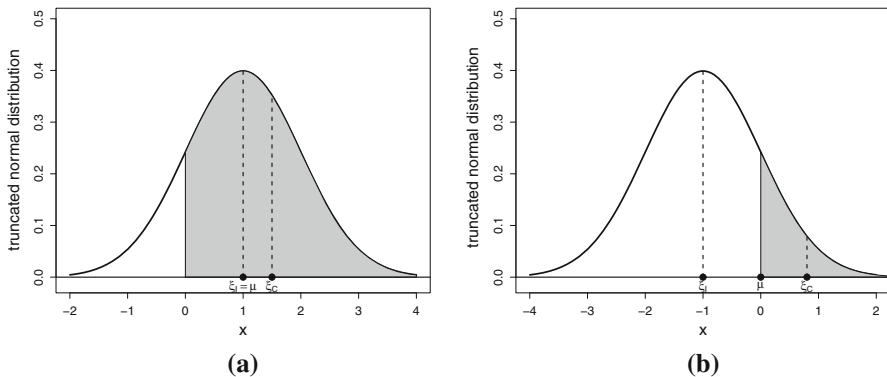


Fig. 1 Two cases of truncated normal variables. The mean is inside (respectively, outside) the acceptance region **(a)** (respectively, **b**)

Remark 4 The inequality mode or MAP (maximum a posteriori) of the conditional GP M_{IK}^N converges uniformly to the constrained interpolation function defined as the solution of the following convex optimization problem

$$\arg \min_{h \in H \cap I \cap C} \|h\|_H^2,$$

where H is a reproducing kernel Hilbert space (RKHS) associated with the positive type kernel K (Aronszajn 1950), I is the set of functions verify interpolation conditions and the convex set C is the space of functions which verify the inequality constraints (Bay et al. 2015, 2016; Maatouk 2015).

This extends to the case of interpolation conditions and inequality constraints the correspondence established by Kimeldorf and Wahba (1970) between Bayesian estimation on stochastic process and smoothing by splines.

In practice, two cases in the simulation of truncated multivariate normal distributions are possible (Fig. 1a, b for example in one dimension). In the first case (Fig. 1a), $\xi_1 = \mu$ and so $m_K^N = M_{\text{IK}}^N$ and they are different from m_{IK}^N . In this case, ξ_1 is inside C_ξ (for instance the nonnegative quadrant) and the usual (unconstrained) kriging mean respects the inequality constraints. The second one, where the three are different (Fig. 1b). In this case, ξ_1 is outside C_ξ and the usual (unconstrained) kriging mean does not respect the inequality constraints.

4 Simulation Study

In this section, the performance of the proposed model in terms of prediction is shown. It is compared with unconstrained GP using cross-validation (CV) methods and maximum likelihood estimator (MLE) in the estimation of the hyper-parameters of covariance functions. The real non-decreasing function $f(x) = -\frac{10^{-2}}{(x+0.2)^3}$ is considered which is evaluated at $X = (0, 0.1, 0.4, 0.6, 0.7, 0.8, 1)$ (black points in Fig. 2a). Predictions at twenty input locations (triangles in Fig. 2a) are compared to the true

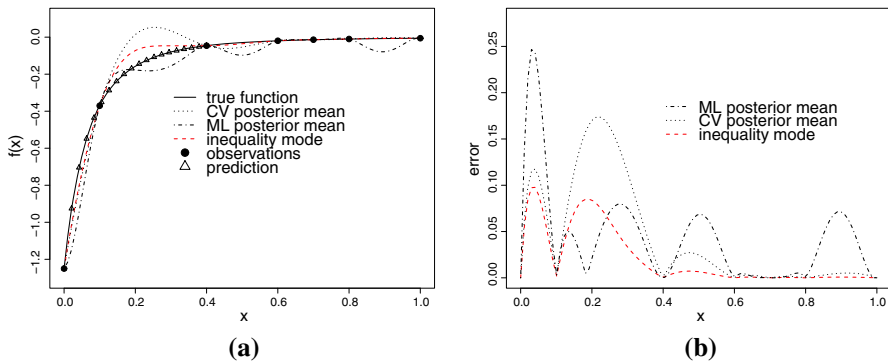


Fig. 2 Comparison in terms of prediction between: posterior maximum of the constrained GP and posterior mean of unconstrained GP (a). Absolute difference between the real function and the estimators (b)

response. These points have been placed in the region of the input space where the real function changes quite rapidly. The Matérn 5/2 covariance function (Table 1) has been used. The dotted line (respectively, dash-dotted line) represents the posterior mean of unconstrained GP where the length parameter $\theta = 0.06$ (respectively, $\theta = 0.11$) is computed by the CV method (respectively, MLE). Let us mention that the R package ‘DiceKriging’ (Roustant et al. 2012) has been used to estimate the length parameter θ of the covariance function. The red line represents the posterior maximum (inequality mode) of the constrained GP (Model 8 conditionally to interpolation conditions and monotonicity (non-decreasing) constraints). The length parameter $\theta = 4$ has been computed by a suited cross-validation method (Maatouk et al. 2015). Firstly, the posterior mean of unconstrained GP using CV and MLE do not respect monotonicity constraints in the entire domain contrarily to the inequality mode. Secondly and more precisely, the Q_2 criterion defined as

$$Q_2 = 1 - \frac{\sum_{i=1}^{20} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{20} (y_i - \bar{y})^2},$$

where y_i , $i = 1, \dots, 20$ are output observations, $\bar{y}$ is the mean of output observations and $\hat{y}_i$ is the value of the prediction function at prediction points. Notice that Q_2 should be as close as possible to one. It is equal to 0.97 when using the inequality mode as an estimator and equal to 0.91 (respectively, 0.88) for the posterior mean of unconstrained GP using MLE (respectively, CV methods). In Fig. 2b, the absolute difference between the real function and the three estimators is shown.

To generalize the numerical result, six design points are randomized one hundred times uniformly on $[0, 1]$. Using the same predictions in the above example, the mean of the Q_2 criterion $\overline{Q_2}$ is computed. It is equal to 0.61 when using the inequality mode as an estimator and 0.54 (respectively, 0.34) for the posterior mean of unconstrained GP as an estimator using CV methods (respectively, MLE) in the estimation of the parameters.

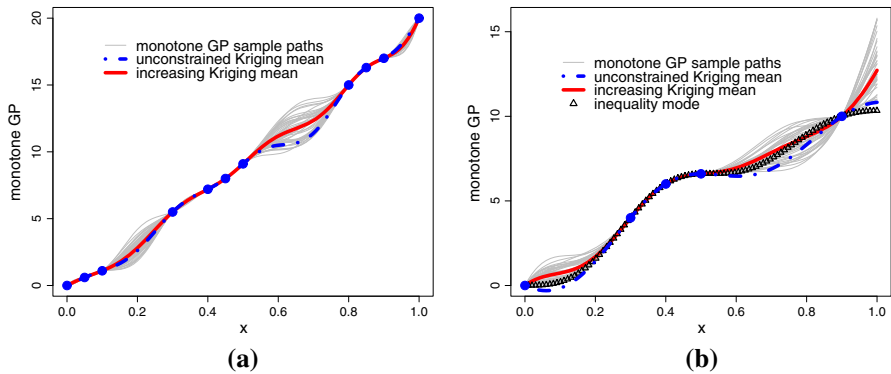


Fig. 3 Simulated paths drawn from Model (8) respecting non-decreasing constraints in the entire domain. The usual kriging mean (*dash-dotted line*) coincides with the inequality mode and respects the monotonicity in **a**, but not in **b**

5 Illustrative Examples

The aim of this section is to illustrate the proposed method with certain constraints such as boundedness, monotonicity and convexity and to show the difference between prediction functions (unconstrained kriging mean, inequality kriging mean and inequality mode). The simulation results are obtained by using Gaussian and Matérn 3/2 covariance functions, where the constrained evaluations are not taken from constrained functions. In one dimension, monotonicity, boundedness and convexity constraints examples are considered. In two dimensions, the monotonicity (non-decreasing) constraints with respect to the two input variables and only one variable are illustrated.

5.1 Monotonicity in One-Dimensional Case

Two monotonicity examples in one dimension (Fig. 3) are considered. In Fig. 3a, the eleven design points are given by $X = (0, 0.05, 0.1, 0.3, 0.4, 0.45, 0.5, 0.8, 0.85, 0.9, 1)$ and the corresponding output $y = (0, 0.6, 1.1, 5.5, 7.2, 8, 9.1, 15, 16.3, 17, 20)$. Let us choose $N = 50$ and generate forty sample paths taken from Model (8) conditionally to given data and monotonicity (non-decreasing) constraints ($Y'(u_j) \geq 0$, $j = 0, \dots, N$). The Gaussian covariance function is used with the hyper-parameters (σ^2, θ) fixed to $(20^2, 0.14)$. Notice that the simulated paths (gray lines) are non-decreasing in the entire domain, as well as the increasing kriging mean m_{IK}^N (solid line). The usual (unconstrained) kriging mean m_K^N and the inequality mode M_{IK}^N (dash-dotted line) coincide and are also non-decreasing. This is because ξ_1 is inside the acceptance region C_ξ . In Fig. 3b, the input is $X = (0, 0.3, 0.4, 0.5, 0.9)$ and the corresponding output is $y = (0, 4, 6, 6.6, 10)$. Again, the Gaussian covariance is used with the parameters (σ^2, θ) fixed to $(20^2, 0.25)$. The increasing kriging mean (solid line) and the inequality mode satisfy monotonicity (non-decreasing) constraints, contrarily to the usual (unconstrained) kriging mean (dash-dotted line): It corresponds to the situation where ξ_1 lies outside the acceptance region C_ξ .

5.2 Boundedness Constraints in One-Dimensional Case

Now, the positive and boundedness constraints are considered (Fig. 4). Suppose $N = 50$ and generate one hundred sample paths taken from the finite-dimensional approximation defined in (6) conditionally to given data and boundedness constraints. In both figures, the Gaussian covariance function is used with the parameters $(\sigma^2, \theta) = (4^2, 0.13)$ (Fig. 4a) and $(\sigma^2, \theta) = (25^2, 0.2)$ (Fig. 4b). In Fig. 4a, ξ_I is inside the acceptance region and the usual (unconstrained) kriging mean coincides with the inequality mode and respects boundedness constraints, contrarily to Fig. 4b, where ξ_I lies outside the acceptance region. Notice that the simulated paths satisfy the inequality constraints in the entire domain [nonnegative (Fig. 4a)] and are bounded between -20 and 20 (Fig. 4b). From Fig. 4b, one can remark that the degree of smoothness of the inequality mode is related to that of the covariance function K of the original GP (Remark 4).

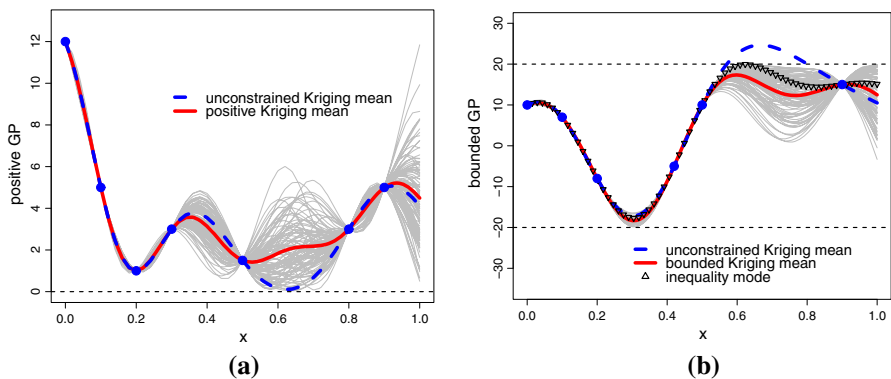


Fig. 4 Simulated paths drawn from Model (6) respecting positivity constraints (a) and boundedness constraints (b). The usual (unconstrained) kriging mean and the inequality mode coincide in a, but not in b

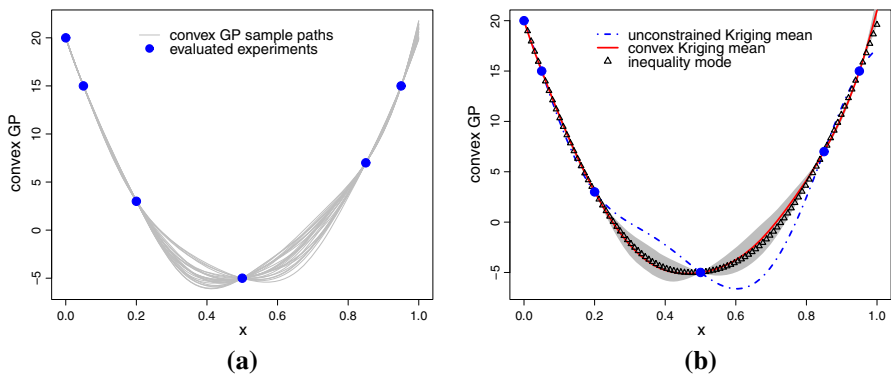


Fig. 5 Simulated paths drawn from Model (9) respecting convexity constraints in the entire domain (a). 95% prediction intervals together with three estimators: unconstrained kriging mean, inequality mode and convex kriging mean (b)

5.3 Convexity Constraints in One-Dimensional Case

The constrained evaluations in Fig. 5 are given by $X = (0, 0.05, 0.2, 0.5, 0.85, 0.95)$ and the corresponding output $y = (20, 15, 3, -5, 7, 15)$. Let us choose $N = 50$ and generate twenty-five sample paths taken from Model (9) conditionally to given data and convexity constraints ($\xi_j \geq 0, j = 0, \dots, N$). The Gaussian covariance function is used with the parameters fixed to $(\sigma^2, \theta) = (10^2, 0.2)$. The simulated paths, the inequality mode (maximum a posteriori) and the convex kriging mean (mean

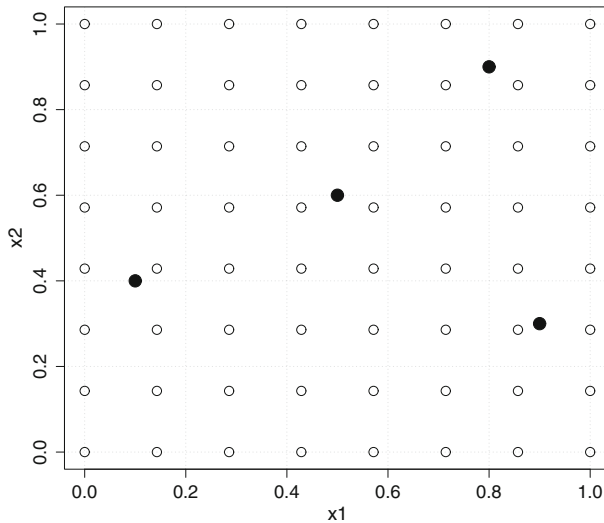


Fig. 6 Design points for the monotone 2D interpolation problem (black points) and knots $(u_i, u_j)_{0 \leq i, j \leq 7}$ used to define the basis functions

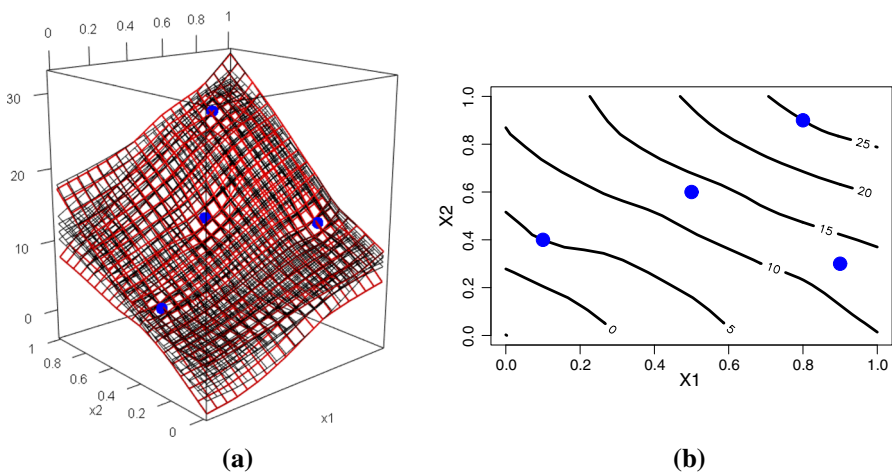


Fig. 7 Simulated surfaces drawn from Model (10) respecting monotonicity constraints for the two input variables (a). Contour levels for one simulated surface (b)

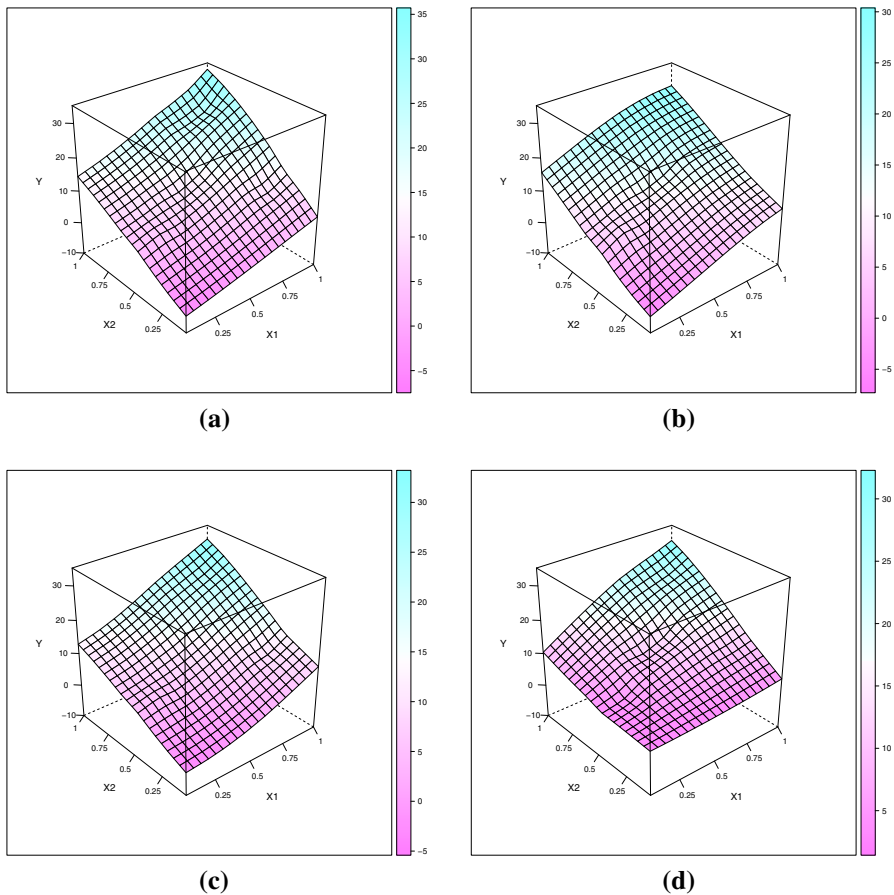


Fig. 8 Simulated surfaces drawn from the example used in Fig. 7a respecting monotonicity constraints for the two input variables

a posteriori) are convex in the entire domain, contrarily to the usual (unconstrained) kriging mean (dash-dotted line). It corresponds to the situation where ξ_1 lies outside the acceptance region C_ξ .

5.4 Isotonicity in Two Dimensions

In two dimensions, the aim is to interpolate a 2D-function defined on $[0, 1]^2$ and non-decreasing with respect to the two inputs. In that case, and by the uniform subdivision of the input set, the number of knots and basis functions is $(N + 1)^2$. In Figs. 6, 7 and 8, $N = 7$, then 64 knots and basis functions derived. Suppose that the real function is evaluated at four design points given by the rows of the 4×2 matrix $X = \begin{bmatrix} 0.1 & 0.9 & 0.5 & 0.8 \\ 0.4 & 0.3 & 0.6 & 0.9 \end{bmatrix}^\top$ and the corresponding output $y = (5, 12, 13, 25)$. The

output values respect monotonicity (non-decreasing) constraints in two dimensions. The two-dimensional Gaussian covariance function is used

$$K(x, x') = \sigma^2 \exp\left(-\frac{(x_1 - x'_1)^2}{2\theta_1^2}\right) \times \exp\left(-\frac{(x_2 - x'_2)^2}{2\theta_2^2}\right),$$

where the variance parameter σ is fixed to 10 and the length parameters (θ_1, θ_2) to $(1, 1)$. Let us generate five simulation surfaces taken from Model (10) conditionally to given data and monotonicity (non-decreasing) constraints with respect to the two input variables (Fig. 7a). The two red surfaces are the 95% prediction interval. To check visually the isotonicity, the contour levels of one simulation surface is plotted

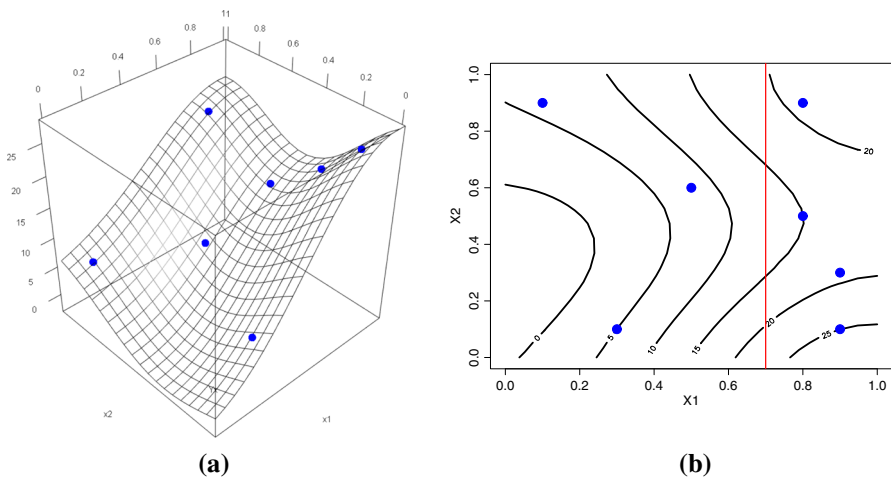


Fig. 9 Simulated surface drawn from Model (11) respecting monotonicity (non-decreasing) constraints for the first variable only, and the associated contour levels

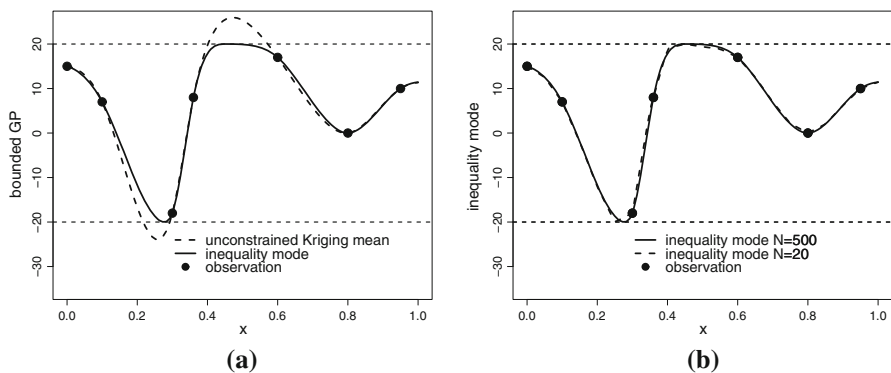


Fig. 10 The solid lines represent the inequality mode M_{IK}^N when $N = 500$ which respect boundedness constraints in the entire domain. The dashed line in **a** represents the unconstrained kriging mean and in **b** the inequality mode when $N = 20$

(Fig. 7b). The blue points represent the interpolation input locations (design points). If one of the variables from which we draw a vertical or horizontal line is fixed, it must not intersect a contour level twice.

In Fig. 8, some simulated surfaces taken from the example used in Fig. 7a are shown. All the simulated surfaces are non-decreasing with respect to the two input variables. In Fig. 9, a simulation surface of the conditional GP at four design points including monotonicity (non-decreasing) constraints with respect to the first input variable only is shown. In that case, suppose, if $N = 23$, then $(N + 1)^2$ basis functions and knots derived. The two-dimensional Gaussian covariance function is used with the variance parameter σ^2 fixed to 10^2 and the length hyper-parameters (θ_1, θ_2) fixed to $(0.5, 0.45)$.

6 Numerical Convergence

In order to investigate the convergence rate of the proposed model when N tends to infinity, the inequality mode and the usual (unconstrained) kriging mean are plotted (Fig. 10a) in the situation where they are different. It corresponds to the case where the usual kriging mean does not respect boundedness constraints (i.e., $\xi_1 \notin C_\xi$). In Fig. 10b, the inequality mode M_{IK}^N of the finite-dimensional approximation defined in (6) when $N = 500$ is plotted. The dashed line represents M_{IK}^N when $N = 20$, which is close to one generated from $N = 500$. Let us specify that the Matérn 3/2 covariance function is used with the length parameter $\theta = 0.25$ (Table 1).

7 Conclusions

A new model for incorporating both interpolation conditions and inequality constraints into a Gaussian process emulator is proposed. The presented method ensures that the inequality constraints are respected not only in a discrete subset of the input set but also in the entire domain. The main idea is based on a finite-dimensional approximation of Gaussian processes which converges uniformly pathwise. It is constructed by incorporating deterministic basis functions and Gaussian random coefficients. The basis functions are chosen such that inequality constraints of Y^N are equivalent to a finite number of constraints on the coefficients. So, the initial problem is equivalent to simulate a Gaussian vector restricted to convex sets. This model has been applied to real data in insurance and finance to estimate a term structure curve and default probabilities (Cousin et al. 2016).

Now, the problem is open to substantial future work. For practical applications, estimating parameters should be investigated and cross-validation techniques can be used. The suited cross-validation method to inequality constraints described in Maa-touk et al. (2015) can be developed. As input dimension increases, the efficiency of the method will become low. Using an uniform grid in dimension d , the size of the Gaussian vector of random coefficients is equal to $(N + 1)^d$, where $1/N$ is the mesh size. It is also equal to the number of basis functions or knots. So the complexity of the approximation model increases exponentially as the dimension d increases. However, the choice of knots (subdivision of the input set) can be improved to reduce the cost

of simulation, as well as the number of basis functions. This problem is also related to the choice of the basis functions with respect to a prior information on the regularity of the real function. Additionally, the simulation of the truncated Gaussian vector can be accelerated by Markov chain Monte Carlo (MCMC) methods or Gibbs sampling (Geweke 1991; Robert 1995).

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Appendix: Proofs of Propositions

Proof (Proofs of Propositions 1 and 2) Let Y^N be defined as in (6) (respectively, in 8). Since $Y(u_j)$, $j = 0, \dots, N$ are Gaussian variables, then Y^N is a GP with dimension equal to $N + 1$ and covariance function

$$\begin{aligned} \text{Cov} \left(Y^N(x), Y^N(x') \right) &= \sum_{i,j=0}^N \text{Cov} \left(Y(u_i), Y(u_j) \right) h_i(x) h_j(x') \\ &= \sum_{i,j=0}^N K(u_i, u_j) h_i(x) h_j(x'). \end{aligned}$$

The derivative of a GP is also a GP, respectively. For all $x, x' \in [0, 1]$,

$$\begin{aligned} K_N(x, x') &= \text{Cov} \left(Y^N(x), Y^N(x') \right) = \text{Var} \left(Y(0) \right) + \sum_{i=0}^N \frac{\partial K}{\partial x} (u_i, 0) \phi_i(x) \\ &\quad + \sum_{j=0}^N \frac{\partial K}{\partial x'} (0, u_j) \phi_j(x') + \sum_{i,j=0}^N \frac{\partial^2 K}{\partial x \partial x'} (u_i, u_j) \phi_i(x) \phi_j(x'). \end{aligned}$$

To prove the pathwise convergence of Y^N to Y , write more explicitly, for any $\omega \in \Omega$

$$Y^N(x; \omega) = \sum_{j=0}^N Y(u_j; \omega) h_j(x).$$

Hence, the sample paths of the approximating process Y^N are piecewise linear approximations of the sample paths of the original process Y . From $h_j \geq 0$ and $\sum_{j=0}^N h_j(x) = 1$, for all $x \in [0, 1]$,

$$\begin{aligned}
|Y^N(x; \omega) - Y(x; \omega)| &= \left| \sum_{j=0}^N (Y(u_j; \omega) - Y(x; \omega)) h_j(x) \right| \\
&\leq \sum_{j=0}^N \sup_{|x-x'| \leq \Delta_N} |Y(x'; \omega) - Y(x; \omega)| h_j(x) \\
&= \sup_{|x-x'| \leq \Delta_N} |Y(x'; \omega) - Y(x; \omega)|. \quad (13)
\end{aligned}$$

By uniform continuity of sample paths of the process Y on the compact interval $[0, 1]$, this last inequality (13) shows that

$$\sup_{x \in [0, 1]} |Y^N(x; \omega) - Y(x; \omega)| \xrightarrow{N \rightarrow +\infty} 0,$$

with probability one. Let us write, respectively, that for any $\omega \in \Omega$

$$Y^N(x; \omega) = Y(0; \omega) + \int_0^x \left(\sum_{j=0}^N Y'(u_j; \omega) h_j(t) \right) dt.$$

From Proposition 1, $\sum_{j=0}^N Y'(u_j; \omega) h_j(x)$ converges uniformly pathwise to $Y'(x; \omega)$ since the realizations of the process are almost surely continuously differentiable. One can conclude that Y^N converges uniformly to Y for almost all $\omega \in \Omega$.

The last property done by the fact that Y^N is a piecewise linear approximation and

$$Y^N(u_i) = \sum_{j=0}^N Y(u_j) h_j(u_i) = \sum_{j=0}^N Y(u_j) \delta_{ij} = Y(u_i) \in [a, b],$$

$i = 0, \dots, N$, (respectively, the fact that the basis functions are non-decreasing and

$$0 \leq (Y^N)'(u_i) = \sum_{j=0}^N Y'(u_j) h_j(u_i) = Y'(u_j),$$

$i = 0, \dots, N$). □

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